

# Advanced Programming with Python (INF 601 / INF 601G)

Fort Hays State University  
Robbins College of Business and Entrepreneurship  
Department of Informatics

## 1. COURSE INFORMATION

Credit hours: 3.0

Prerequisite(s): INF 360: Programming with Python

Semester and year: Fall 2026

Location of class: Campus / Online

Class time: 1:30 p.m. – 2:45 p.m., Tuesday and Thursday (section A). Online sections are asynchronous; all material is in Blackboard.

Live sessions: joined by Zoom. The meeting link is posted in Blackboard under “Live Zoom Meeting Link” (Meeting ID 880 4563 7212). The link is for enrolled students only — please do not share it outside the course.

Virtual course: Material will be available in Blackboard

### **Key semester dates (FHSU Registrar, Fall 2026)**

17 August — Classes begin

30 August — Final day to add or register in a full-semester course; final day for 100% refund

7 September — Labor Day holiday, University closed; classes resume 8 September

14 September — Final day to withdraw from a course without a transcript record (20th day of classes)

25 October — Final day to withdraw with a transcript record of “W”. No withdrawals allowed after this date.

23–27 November — Fall Break (University closed 26–27 November); classes resume 30 November

5, 7–11 December — Final examinations

11 December — Semester ends

*Dates above are from the University's published Fall 2026 academic calendar and govern if anything elsewhere in this syllabus disagrees: <https://www.fhsu.edu/registrar/fall-semester-2026-academic-calendar>*

## 2. INSTRUCTOR INFORMATION

Jason Zeller, Assistant Professor

HHM251  
jlzeller@fhsu.edu  
785-628-4758

Course communication: the course Discord server is the main communication path for questions, debugging help, and day-to-day conversation; the invite is in Blackboard. Anything concerning your grade or personal circumstances should come to me by Blackboard message or email rather than a public channel.

### 3. TEXTBOOK AND COURSE MATERIAL

Required textbooks: none. Open Educational Resources (OERs) provided in Blackboard.

Supplementary book(s) and article(s): Supplementary Learning Resources provided in Blackboard.

Technology requirement: a computer (PC/Mac/Linux) with a terminal, supporting web browsers and text editors. Bring Your Own Device. No specific IDE is required.

#### **Claude Code subscription required.**

This course uses Claude Code (Anthropic's agentic coding tool) as a core development harness. Students must have an active Claude subscription that includes Claude Code and have it installed by the end of Week 1. The Claude Pro plan — \$20 per month billed monthly, or about \$17 per month billed annually — includes Claude Code and is all this course requires; a Max plan is not necessary. The free tier does not include Claude Code. See <https://claude.com/pricing>.

#### **Course Practice Hub.**

A course-hosted REST API at <https://practice.fhsucyber.com> that you will call from Week 2 onward, use as a data source for the NumPy and Pandas projects, and finally wrap as your own MCP server in Week 13. Register for an account and save your API token in Week 1.

### 4. COURSE DESCRIPTION

In this project-based class, students who are already familiar with Python will learn more advanced Python programming and techniques. Students will learn how to utilize both native and third-party Python libraries, robust development tools, and current frameworks to develop efficient solutions to challenges often found in the computing field.

This offering additionally treats agentic development as a first-class skill: students work throughout the semester in an AI coding harness, learn to engineer the agent loop (verification, termination, and recovery), build a Model Context Protocol (MCP) server over a live REST API, and author an Agent Skill.

## 5. COURSE LEARNER OUTCOMES

### **UNDERGRADUATE**

Course Learner Outcomes:

1. Identify fundamental Python programming concepts
2. Understand terminal-based development and the use of an AI coding harness
3. Apply appropriate frameworks and libraries
4. Assess the Object-Oriented Programming approach to development
5. Evaluate syntax, operators, expressions, methods, and flow control
6. Create functional applications using Python
7. Use an agentic AI coding harness (Claude Code) responsibly to design, build, and debug Python software, and explain the code produced — including engineering the agent loop with an objective feedback signal (tests), a testable termination condition, and a stopping/escalation rule
8. Build a Model Context Protocol (MCP) server that exposes a REST API as tools, and author an Agent Skill that drives it

### **GRADUATE LEVEL**

Course Learner Outcomes:

1. Identify fundamental Python programming concepts
2. Understand terminal-based development and the use of an AI coding harness
3. Apply appropriate frameworks and libraries
4. Assess the Object-Oriented Programming approach to development and express the importance of coding style compliance
5. Evaluate syntax, operators, expressions, methods, and flow control
6. Create functional solutions using Python to address existing needs
7. Use an agentic AI coding harness (Claude Code) responsibly to design, build, and debug Python software, and explain the code produced — including engineering the agent loop with an objective feedback signal (tests), a testable termination condition, and a stopping/escalation rule
8. Build a Model Context Protocol (MCP) server that exposes a REST API as tools, and author an Agent Skill that drives it

## 6. TEACHING, LEARNING METHODS, & COURSE STRUCTURE

Delivery Method: lectures, labs, videos, readings, quizzes, problem sets, and projects

Course Structure:

1. Review of Python Basics

2. Terminal-based development and the Claude Code harness
3. Object-Oriented Programming Review and File Organization
4. Introduction to NumPy and SciPy
5. Understanding Dataframes
6. Web Frameworks Based on Python
7. Dashboard Visualizations for Data
8. Python for Cybersecurity and Cryptography
9. Agentic development and loop engineering: the agent loop, harnesses, prompts-as-programs, and engineering the loop (verification, termination, recovery)
10. Model Context Protocol (MCP) servers and Agent Skills

## 7. COURSE SCHEDULE

This schedule is tentative and might change during the semester depending on how the course evolves. The content is subject to change depending on students' interest and progress. Students will be notified of changes through announcements in class or at the Blackboard course site.

Unless stated otherwise, work is due Sunday at 11:59 p.m. Central of its listed week. There is no instructional week during Fall Break (23–29 November); Week 15 is the week of 30 November and Week 16 is finals week.

| Week | Content                                                                                       | Graded                             |
|------|-----------------------------------------------------------------------------------------------|------------------------------------|
| 1    | Git/GitHub, Python 3.13, Claude Code install & terminal fundamentals; Practice Hub onboarding | —                                  |
| 2    | OOP review, file organization, REST fundamentals against the Practice Hub                     | —                                  |
| 3    | API CRUD lab (requests + an OOP client class)                                                 | MP1: API Client                    |
| 4    | NumPy / SciPy / Matplotlib (Claude Code integrated)                                           | Quiz 1; MP2; Check-In Bot assigned |
| 5    | Pandas dataframes (Claude Code integrated)                                                    | Quiz 2; MP3                        |
| 6    | Streamlit dashboards (read from the Practice API)                                             | Check-In Bot due                   |
| 7    | Agentic foundations & loop engineering: agent loop, harness, CLAUDE.md, plan                  | Quiz 3; MP4 assigned               |

|    |                                                                                                                              |                                  |
|----|------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
|    | mode;<br>prompt → context → harness → loop; verification, termination, recovery                                              |                                  |
| 8  | MP4 concentration-split agentic project; Flask begins                                                                        | MP4 due                          |
| 9  | Flask (templates, SQLite, auth, Bootstrap)                                                                                   | Quiz 4; MP5                      |
| 10 | Django + admin                                                                                                               | Midterm Exam                     |
| 11 | Django (continued); freeze sites as pentest targets                                                                          | MP6                              |
| 12 | Build & Break: authorized Pentest Week (cyber pentest peers' Django apps over the course VPN; CS/other defend and remediate) | Pentest Week discussion (graded) |
| 13 | MCP deep-dive: build an MCP server over the Practice API                                                                     | MP7; Final Project outline       |
| 14 | Agent Skills, harness customization, and the Claude Agent SDK lab                                                            | Quiz 5; Hermes Lab assigned      |
| 15 | Final project — framework build                                                                                              | Hermes Lab due                   |
| 16 | Final project — debugging, submission, instructor review + live agent demo                                                   | Final submission; Review         |

## 8. ASSESSMENT METHODS AND GRADING SCALE

This course has a weighted total score.

### Module Quizzes: 20%

Five quizzes given throughout the semester: NumPy, Pandas, agentic concepts and loop engineering, Flask, and MCP & Agent Skills. Quizzes are closed-AI.

### Mini Projects & Labs: 25%

Seven mini projects plus two standalone graded labs — the Scheduled Check-In Bot (a GitHub Actions automation run against the Practice Hub, Weeks 4–6) and the Hermes Lab (your own agent on the course server, Weeks 14–15). Each of the nine is graded out of 100 points and

includes an “## AI Usage” section in its README. All nine are weighted equally within the category — each is 1/9 of the 25%, roughly 2.8% of the final grade — so the category mark is the average of the nine 100-point scores.

**Graded Discussion — Pentest Week (Week 12): 5%**

A Build & Break activity in which cybersecurity students conduct an authorized penetration test of classmates' Week-11 Django applications over the course VPN, while computer science and other students defend and remediate their own. It is graded as a discussion and is not one of the nine Mini Projects & Labs. Authorization is scope-limited by the course VPN and a signed Rules of Engagement; testing out-of-scope systems is an academic honesty violation under §11.1. Participation is required. It is graded out of 100 points against the rubric on the assignment page and forms its own 5% category.

**Midterm Exam: 20%**

An exam assessing skills and topics covered to this point in the semester, focusing on object-oriented programming. Closed-AI. Graduate students additionally complete a short standards essay.

**Concentration Specific Final Project: 30%**

A project allowing students to apply the skills covered during the semester, plus an agentic requirement: expose part of the project via an MCP server or author an Agent Skill used during development, demonstrated live at the Week 16 review. Graduate projects require both an MCP server and an Agent Skill, in addition to the existing public-deployment requirement.

**Grading Scale:**

90% - 100%: A (High Honors) outstanding and distinguished meeting of course objectives.

Excellent understanding and appreciation for the depth and breadth of the subject matter.

80% - 89%: B (Honors) highly satisfactory and successful meeting of course objectives.

Superior interest and understanding of the subject matter.

70% - 79%: C (Satisfactory) successful and respectable meeting of course objectives. Work is usually completed to meet minimum requirements of assignments.

60% - 69%: D (Low Level Passing) work sometimes fails to meet minimum requirements of assignments and often is below the quality generally considered acceptable by an employer.

<59%: U (Failing) failure to meet minimum course requirements.

## 9. STUDENT HELP RESOURCES

Students have access to academic services, technical support and student services at Fort Hays State University. You can find the resources online at <http://www.fhsu.edu/ctelt/services/Student-Help-Resources/>

## 10. COURSE POLICIES

Disclaimer: the Instructor reserves the right to make course changes if necessary, and will notify students of these changes through Blackboard announcements and e-mail.

Course material: all course content is available on Blackboard.

Class attendance: on-campus students are expected to attend classes.

Assignment due date: work is due Sunday at 11:59 p.m. Central of its listed week. Makeup (late) work is not accepted unless PRIOR approval has been granted by the instructor.

Procedures for assignment submission: mini projects and labs are submitted as GitHub repositories following the naming convention given in each assignment; everything else is submitted via Blackboard unless differently notified.

Bonus points: bonus points are granted to students who participate in research projects and activities connected to the course.

### 10.1 Use of AI — “AI-assisted, you own it.”

Claude Code and other AI tools are permitted on all coursework from Week 1, because using them well is part of what this course teaches. However:

1. You own every line you submit. You must understand your code and be able to explain any part of it on request — in quizzes, the midterm, and the Week 16 review meeting. “The AI wrote it” is not an acceptable explanation.
2. Document your AI use. Every mini project, standalone lab, and the final project README must include a short “## AI Usage” section: which parts were AI-assisted, which you wrote by hand, and what you changed in the AI's output. For the agent-driven labs, also account for what the agent did versus what you decided, prompted, and corrected.
3. Quizzes and the midterm are closed-AI unless stated otherwise. They exist to confirm individual understanding.
4. Academic honesty still applies. Submitting work you cannot explain, or misrepresenting authorship, is a violation under §11.1.

## 11. UNIVERSITY POLICIES

### Tiger pact

I am a tiger. I belong to a strong unique family who strives for greatness and success. I instill integrity and confidence within others as well as myself. I incorporate honesty and responsibility in all that I do. I am the future.

### 11.1 Academic Honesty Policy

Membership in the FHSU learning community imposes upon the student a variety of commitments, obligations and responsibilities. It is the policy of FHSU to impose sanctions on students who misrepresent their academic work. These sanctions will be selected by appropriate classroom instructors or other designated persons consistent with the seriousness

of the violation and related considerations... Students participating in any violation of this policy must accept the consequences of their actions. Classroom instructors and/or university review/appeals committees and administrators will assess the sanctions for violation of this policy.

### **11.2 Statement of Accessibility and Services for Students with Disabilities**

If you have a disability that may have an impact on your ability to carry out assigned course work and if you wish to seek any accommodations for this course, you must contact Services for Students with Disabilities (SSD). SSD is located in the Kelly Center, Picken Hall, Room 111, 785-628-4401. SSD will review your documentation and determine, with you, what academic accommodations are necessary and appropriate for you that can be accommodated in this course. All information and documentation of your disability is confidential and will not be released by SSD without your written permission. Students can find more information at <http://www.fhsu.edu/disability/get-access/>. Instructors who need help to create instructional materials for students with special needs can seek help from Learning Technologies (LT), 785-628-4194.

### **11.3 Title IX**

FHSU is committed to fostering a safe and productive learning environment. Title IX makes it clear that violence and harassment based on sex, gender, and gender identity are Civil Rights offenses subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, etc. This includes all types of gender and relationship violence, sexual harassment, sexual misconduct, domestic and dating violence, and stalking. If you wish to report an incident or have questions about school policies and procedures regarding Title IX issues, please contact Amy Schaffer, University Compliance Officer and the FHSU Title IX Coordinator, at [alschaffer@fhsu.edu](mailto:alschaffer@fhsu.edu) or (785) 628-4175. The Compliance Officer can help connect you to campus and outside resources, discuss all of your reporting options, and assist with any concerns you may have.

### **11.4 Concealed Carry Policy**

Under the Concealed Carry Policy, individuals who carry a concealed handgun must have the handgun on or about their person at all times. Backpacks are appropriate for carrying a handgun as long as the backpack remains within the exclusive and uninterrupted control of the individual. A backpack or other bag used to carry a handgun must be within the immediate reach of the individual. In this course, the instructor may require students to place backpacks or other bags out of reach. The instructor will notify students in advance of this requirement so a student who carries a concealed handgun in a backpack or a bag may take steps to conceal the handgun on his or her person before arriving or otherwise plan accordingly. Alternatively, this course involves a lab where metal and/or explosive devices (ammunition) are prohibited by Federal Regulations due to the nature of the lab and for safety concerns. Concealed carry handguns will need to be properly stored prior to such labs. More information related to Concealed Carry can be found at <http://www.fhsu.edu/news/Fort-Hays-State-posts-policy-on-compliance-with-law-allowing-concealed-carry-of-handguns/>

### **11.5 Use of Computing Resources**

Fort Hays State University (FHSU) provides computing resources and worldwide network access to its faculty, staff, and students for legitimate administrative, educational, and research efforts. As a member of the FHSU electronic community it is your responsibility to use computing resources ethically and responsibly. Members of the FHSU electronic community are expected to use computing resources ethically, and to exercise reasonable care in utilization of FHSU information systems or their components. More information related to privacy, responsibilities, things forbidden to do and use of email can be found at [http://www.fhsu.edu/academic/provost/handbook/ch\\_1\\_computing\\_resources/](http://www.fhsu.edu/academic/provost/handbook/ch_1_computing_resources/)

### **11.6 Withdrawal Policy**

Students may withdraw full -semester courses through 11:59:59PM CT on the 35th day of the semester (Learning Technologies (LT) will work with the Registrar's Office and Technologies Services (TS) to make the specific date for each semester available at the syllabus site). Students withdrawing during this time period will not receive any notation on their transcript. Students who withdraw after this period and thru 11:59:59PM CT on the 70th day of the semester will receive a notation on the transcript of withdrawal (W). No withdrawals after the 70th (LT will work with the Registrar's Office and TS to make the specific date for each semester available at the syllabus site) day of the semester. Students who withdraw completely will receive a notation on their transcript of the date withdrawn. Students receiving financial aid have additional responsibility and should contact the Office of Student Financial Assistance, 785-628- 4408. (<http://www.fhsu.edu/registrar/academic-policies-and-information/>)